Hazem A. F. Younis

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Education:

2027	**PhD**	Computer Engineering, expected May 2027	The University of Texas at Dallas
2022	**MS**	Computer Engineering	The University of Texas at Dallas
2019	**BS**	Electrical Engineering	University of Wisconsin–Platteville

Graduate coursework: Advanced Digital Logic; Computer Architecture; Microprocessor Systems; Real-Time Systems; Design and Analysis of Reconfigurable Systems; Digital Signal Processing

Teaching Interests:

Digital logic and digital systems; computer architecture; microprocessors and embedded systems; electric circuits; Verilog and FPGA laboratories; introductory ECE and first-year engineering

Teaching Experience:

- **2026**
  - Introduction to Engineering and Computer Science, ECS 1100, Fall, instructor of record, 21 students
  - Social Issues and Ethics in Engineering, CE 3161, Summer, online, teaching assistant
  - Introduction to Electrical and Computer Engineering II, EE 1202, Spring, teaching assistant
  - Introduction to Digital Systems, CE 2310, Spring, teaching assistant
- **2025**
  - Introduction to Engineering and Computer Science, ECS 1100, two lectures, Fall, instructor of record, 493 students
- **2024**
  - Introduction to Engineering and Computer Science, ECS 1100, Fall, instructor of record, 11 students
- **2017–18**
  - Digital Circuit Design, Microprocessor Systems, and Analog Circuit Design, University of Wisconsin–Platteville, undergraduate teaching assistant

Teaching Effectiveness:

- Students rated “Overall, this instructor was excellent” at 4.60/5.00 across two Fall 2025 lectures of 493 students; 94% agreed or strongly agreed
- Students rated “The instructor evaluated students fairly” at 4.66/5.00; 98% agreed or strongly agreed
- Formal teaching observations for the UT Dallas Graduate Teaching Certificate noted a welcoming, approachable classroom
- In Fall 2025 mid-semester feedback, students most often praised clear expectations, engaging class sessions, and a responsive, approachable instructor

Work Related Experience:

The University of Texas at Dallas ***01/2020–Current***

Instructor of Record, ECS 1100 08/2024–Current

Introduction to Engineering and Computer Science, First Year Program Richardson, TX

- Teach a required first-year course to 525 students: engineering design in 2024–2025, then professional skills, ethics, and industry sectors in 2026
- Plan lessons, lecture, design assessments, grade, and advise students
- Meet weekly with course faculty to reflect on each session and help redesign the course for future semesters
- Co-developed lecture decks, assignments, in-class activities, and video recordings used across all course sections
- Scaled materials and assessments for two Fall 2025 lectures of 297 and 196 students
- Designed an estimation and Excel modeling session built on a live demonstration and a group estimation activity
- Adapted an AI ethics session around real cases of data bias and a five-role structured debate
- Led industry sector weeks with student panels on semiconductors and medical devices
- Assessed learning with before-and-after concept maps that showed how industry panels deepened students’ view of engineering and computing fields

- Use concept checks, “what-if” prompts, and peer “10-minute talks” to keep large classes active
- Used Connections assignments to build student engagement and confidence

Graduate Teaching Assistant 01/2026–08/2026
Department of Electrical and Computer Engineering Richardson, TX

- Led EE 1202 laboratory sections on RC, RL, and op-amp circuits
- Graded EE 1202 quizzes, assignments, and lab reports
- Taught a debugging routine: confirm power, probe stage by stage, and check every input, switch, and output
- Led CE 2310 lab sections: explained digital logic concepts, guided students through building and debugging adders and multiplexers, and graded their work
- Reviewed discussion posts and proctored exams for CE 3161, an online engineering ethics course

Graduate Research Assistant 01/2020–Current
CRSS Cochlear Implant Lab, Center for Robust Speech Systems Richardson, TX

- Build deep-learning voice-conversion models that make unfamiliar talkers sound closer to voices cochlear implant users know, to test whether that improves understanding
- Design and run listening studies comparing cochlear implant and normal-hearing listeners
- Architected CCi-CLOUD, an AWS platform for secure, low-latency remote testing with cochlear implant users
- Contributed to the NIH-funded CCi-MOBILE project, R01 DC016839, about $700,000, and to NIH grant proposals
- Lead an undergraduate project on slowing instructional speech for older listeners while keeping lip sync

Graduate Research Assistant, Texas Instruments–Sponsored Project 01/2020–08/2021
In collaboration with Texas Instruments Dallas, TX

- Deployed real-time speech-command recognition on Raspberry Pi hardware, reaching 98.8% peak accuracy
- Worked on site with TI engineers in design reviews, requirements planning, and project presentations
- Integrated a 4-microphone array and programmable LEDs for on-device feedback on detected commands

Seagrave Fire Apparatus ***01/2018–08/2018***

Electrical Engineering Co-op Clintonville, WI

- Managed the J1939 CAN-bus transition from 250K to 500K baud with OEM suppliers
- Programmed vehicle lighting modules in IQAN Design; designed equipment brackets in SolidWorks and wiring harness layouts in AutoCAD
- Wrote Engineering Change Notices and worked with installation crews to apply them

University of Wisconsin–Platteville ***2016–2018***

Undergraduate Teaching Assistant 2017–2018
Department of Electrical Engineering Platteville, WI

- Taught HCS12 assembly programming in the microprocessor lab, checking student code and debugging hardware
- Mentored op-amp and transistor circuit design projects

Academic Tutor 2016–2017
Student Success Center Platteville, WI

- Tutored first-generation students in calculus, circuits, signals and systems, and linear algebra
- Led exam review sessions; one student I tutored raised his exam scores by 25%

Design Projects:

- *Microcoded CPU.* 8-bit CPU with microsequencer control, built on a four-person team: Verilog datapath, horizontal microcode, and an eight-instruction set with stack push and pull; verified in Quartus II simulation and demonstrated on an Altera DE2 FPGA board
- *FPGA video game.* Maze-chase game built on a five-person team for an Altera DE2 FPGA board: 640×480 VGA output, a character ROM loaded from a memory file, and switch and push-button controls, written in VHDL and Verilog in Quartus II
- *Railway crossing controller.* Mealy state machine in Verilog with an asynchronous reset, verified with a self-written testbench across all train positions

- *Temperature measurement system.* RTD sensor signal conditioning with National Instruments DAQ hardware and a LabVIEW interface for display, logging, and calibration

Courses Prepared to Teach:

Digital Logic and Digital Systems Design; Advanced Digital Design with Hardware Description Languages; Computer Architecture; Microprocessors and Embedded Systems; Electric Circuits; Introduction to Electrical and Computer Engineering; Senior Design

Guest Lectures and Invited Talks:

- Guest Lecturer, EESC 6366 Speech and Speaker Recognition, graduate course, UT Dallas, 2026
- Panelist and co-planner, "Grad School Myths, Told Through Our Stories," UT Dallas Living Learning Communities, 2026
- Research panelist, ECS 1100 Health Innovation and Medical Devices panel with Edwards Lifesciences, Codan, and Vizient, UT Dallas, 2026
- Invited speaker, Battle of the Brains High School Programming Contest, about 200 students, UT Dallas, 2024

Mentoring and Outreach:

- Undergraduate Research Mentor, CRSS Cochlear Implant Lab, UT Dallas, 2026–current: designed and supervise a sophomore's semester project on audiovisual speech-rate adaptation for older cochlear implant listeners
- Education Mentor, Short-term Working Groups on student engagement and disability accommodations, UT Dallas, 2024–2025
- Senior Design Mentor, team that built a mobile audio-crowdsourcing app for lab research data
- STEM Outreach Volunteer, "Kids Make STEM With Heart" workshops, IEEE Dallas Joint Chapter, 2026
- Emcee and organizer, International Night, International Club, UW–Platteville
- Member, International Student Advisory Board, UW–Platteville

Additional Training/Certifications:

- *Graduate Teaching Certificate*; UT Dallas Center for Teaching and Learning, 2025
- *Engineers Who Teach Professional Badge*; Erik Jonsson School, UT Dallas, 2024
- *Teaching associate training* and active-learning seminars; Erik Jonsson School, UT Dallas, 2024–2026
- *Verilog Language and Application*; Cadence, 2020
- *SystemC Language Fundamentals*; Cadence, 2020

Professional Affiliations:

- IEEE Dallas IAS/PES/PELS Joint Chapter, 2025–current
- American Cochlear Implant Alliance, 2023–2025
- Acoustical Society of America, 2021–2022

Publications:

Journal Manuscripts:

1) Younis, H. A. F., Saba, J. N., and Hansen, J. H. L. "CCi-CLOUD: A cloud-based framework for remote auditory testing with cochlear implant users." Submitted to Cochlear Implants International, 2026.

Peer Reviewed Conference Papers:

1) Younis, H., and Hansen, J. H. L. "Quantifying talker familiarity benefits in cochlear implant user speech perception: A baseline solution for generative voice conversion." Accepted to IEEE Spoken Language Technology Workshop, 2026.
2) Hansen, J. H. L., Saba, J. N., Younis, H. A. F., Lawson, T., Okei, S. O., and Chiang, H.-T. "Assessing speech processing HA/CI advancements for naturalistic field testing: Advancements with CCi-MOBILE research platform." Proc. Forum Acusticum, 11th Convention of the European Acoustics Association, Málaga, Spain, 2025, pp. 5213–5218.
3) Younis, H., and Hansen, J. H. L. "Challenges in real-time embedded IoT command recognition." Proc. IEEE 7th World Forum on Internet of Things, 2021, pp. 848–851.

Conference Presentations:

1) Younis, H., and Hansen, J. H. L. “Speaker-in-the-loop: The role of recognizable voices in cochlear implant speech intelligibility.” Poster, Conference on Implantable Auditory Prostheses, Lake Tahoe, CA, July 2025.
2) Younis, H., and Hansen, J. H. L. “CCi-CLOUD: A framework for community based remote cochlear implant user experiments based on the CCi-MOBILE research platform.” Poster, Conference on Implantable Auditory Prostheses, Lake Tahoe, CA, July 2023; and 183rd Meeting of the Acoustical Society of America, Nashville, TN, Dec. 2022, J. Acoust. Soc. Am. 152.4 (2022): A141.
3) Younis, H. A. F., Radheshyam, P., and Hansen, J. H. L. “Leveraging cloud computing resources to enhance CCi-MOBILE functionality.” Poster, Conference on Implantable Auditory Prostheses, Lake Tahoe, CA, 2021, p. 217.

Technical Skills:

Digital and embedded: Verilog, VHDL, SystemC, Xilinx Vivado and Vivado HLS, Intel Quartus, HCS12 assembly, C/C++, Raspberry Pi, ESP32, LabVIEW, NI DAQ, J1939 CAN bus

Programming and tools: Python, MATLAB, AWS, AutoCAD, SolidWorks

Speech and machine learning: PyTorch, TensorFlow Lite; XVC and StarGANv2 voice conversion; ACE cochlear implant coding strategy